



Stony Brook University

DATA ANALYSIS

AMS 572

Inference on one population mean μ

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One population setting

Cross-sectional study: Collect data to test a hypothesis about the mean or median of X

- ▶ Take a random sample from the population.
- ▶ There are different types of sampling schemes. The simplest is the simple random sampling in which every subject in the population has equal chance to be selected.

Statistical inference on one population mean

Suppose we have a random sample of size n : $X_1, X_2, \dots, X_n$ and we wish to draw inference about the population mean μ .

1. Point estimator

- ▶ $\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n} = \frac{\sum_{i=1}^n X_i}{n}$

- ▶ Other estimators: median, mode, trimmed mean

2. Confidence Interval (C.I.)

3. Hypothesis Test

- ▶ Example: μ is the height of adult US males

$$H_0 : \mu \leq 5'6'' \text{ vs } H_1 : \mu > 5'6''$$

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Recap: Normal Distribution

- Probability Density Function (p.d.f.)

X follows normal distribution of mean μ and variance σ^2

$$X \sim N(\mu, \sigma^2)$$

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, -\infty < x < \infty, x \in R$$

$P(a \leq X \leq b) = \int_a^b f(x)dx$ = area under the pdf curve
bounded by a and b

- Cumulative Density Function (c.d.f.)

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(t)dt$$

$$f(x) = [F(x)]' = \frac{d}{dx}F(x)$$

Recap: Normal Distribution

► Standard Normal Distribution

$$Z \sim N(0, 1)$$

$$X \sim N(\mu, \sigma^2)$$

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

$$\begin{aligned} P(a \leq X \leq b) &= P\left(\frac{a - \mu}{\sigma} \leq \frac{X - \mu}{\sigma} \leq \frac{b - \mu}{\sigma}\right) \\ &= P\left(\frac{a - \mu}{\sigma} \leq Z \leq \frac{b - \mu}{\sigma}\right) \end{aligned}$$

Also known as **Z-score**

Theorem: If two random variables have the same MGF, they have the same distribution.

Normal Distributions: $X \sim N(\mu, \sigma^2)$

$$M_X(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$$

Example: If the mgf of W is $e^{-7t+10t^2}$, what is the distribution of W ?

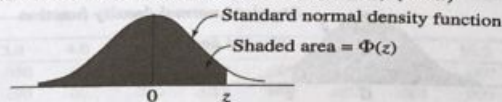


Linear transformation of a normal random variable

► If $X \sim N(\mu, \sigma^2)$ and $Y = aX + b$, a, b are constants, then

$$Y \sim N(a\mu + b, a^2 \sigma^2)$$

Example: Suppose the height of students of AMS 572 is normally distributed. Ten percent of the students are over 6.5 feet tall, while the variance is 0.390625 (or 0.625^2). What is the probability that the height of a student is in between 6 and 7 feet?

Table A.3 Standard Normal Curve Areas $\Phi(z) = P(Z \leq z)$ (cont.)

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9278	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

Distribution of sample mean $\bar{X}$

Theorem: Let $X_1, X_2, \dots, X_n$ be a random sample from a normal population with mean μ , variance σ^2 . Then, the distribution of $\bar{X}$ is

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$



Inference for μ when σ^2 is known under normal distribution



Inference for μ when σ^2 is known under normal distribution

Setup:

- ▶ Assume that the distribution is normal
- ▶ Let $X_1, X_2, \dots, X_n$ be a random sample for a normal distribution with mean μ and variance σ^2 . That is,
 $X \stackrel{iid.}{\sim} N(\mu, \sigma^2), i = 1, \dots, n.$
- ▶ Assume that σ^2 is known.

Inference for μ when σ^2 is known under normal distribution

1. Point estimator for μ

$$\hat{\mu} = \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Note that $E(\hat{\mu}) = E(\bar{X}) = \mu \Rightarrow \hat{\mu} = \bar{X}$ is an unbiased estimator of μ

- $\bar{X}$ is also a maximum likelihood estimator (MLE) and method of moment estimator (MME) of μ .



Inference for μ when σ^2 is known under normal distribution

2 Confidence Interval for μ :

Intuitive approach : $P(c_1 \leq \mu \leq c_2) = 0.95$

$$P(\bar{X} - c_1 \geq \bar{X} - \mu \geq \bar{X} - c_2) = 0.95$$

$$\bar{X} - \mu \sim N(0, \frac{\sigma^2}{n})$$

$$P(\frac{\bar{X} - c_1}{\sigma/\sqrt{n}} \geq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \geq \frac{\bar{X} - c_2}{\sigma/\sqrt{n}}) = 0.95$$

There ARE many ways to choose the c 's.

Pivotal quantity with symmetric pdf, the symmetric CIs are optimal, i.e., they have the shortest lengths for a given confidence level.



Pivotal Quantity (P.Q.) approach for deriving confidence interval

Definition: A pivotal quantity is a function of the sample and parameter of interest whose probability distribution does not depend on the unknown parameters.

Pivotal Quantity (P.Q.) approach for deriving confidence interval

Consider $\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$, where σ^2 is known

- ▶ Is $\bar{X}$ a pivotal quantity for μ ?
- ▶ Function of $\bar{X}$ and μ : $\bar{X} - \mu \sim N(0, \frac{\sigma^2}{n})$
- ▶ Another function of $\bar{X}$ and μ : $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$

Derivation for the symmetrical CI's for μ based on pivotal quantity Z

CI for μ , $0 < \alpha < 1$ (e.g. $\alpha = 0.05 \Rightarrow 95\%$ C.I.)

$$P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) = 1 - \alpha$$

$$P(-z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}) = 1 - \alpha$$

$$P(-z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \leq \bar{X} - \mu \leq z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}) = 1 - \alpha$$

$$P(\bar{X} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}) = 1 - \alpha$$

Thus the $100(1 - \alpha)\%$ C.I. for μ is $[\bar{X} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}]$